

Experiment 14 Robot Framework Data Driven Testing

Overview

This experiment shows how Robot Framework can reuse one login workflow with several sets of credentials. Variables hold shared configuration, while a test template applies the same keyword sequence to every row in the test-case table. The result is less duplicated automation code and simpler test expansion.

Aim

Build an invalid-login test for the SauceDemo website and execute it with three different username and password combinations.

Tools and concepts

- Robot Framework
- SeleniumLibrary
- Robot Framework test templates
- The `*** Variables ***` section
- Keyword arguments using `[Arguments]`
- Data-driven test execution

How the implementation works

The URL and browser name are declared once as suite variables. The Test Template setting points to Invalid Login Scenario, so each row under Test Cases supplies the two arguments expected by that keyword. The keyword opens the site, enters the supplied credentials, submits the form, verifies that the error container appears, and closes the browser through teardown.

The original handout refers to an external data file, but the supplied code stores its data in the Robot Framework test-case table. This rewritten implementation keeps that behavior so it matches the provided experiment and runs without an additional data-file parser.

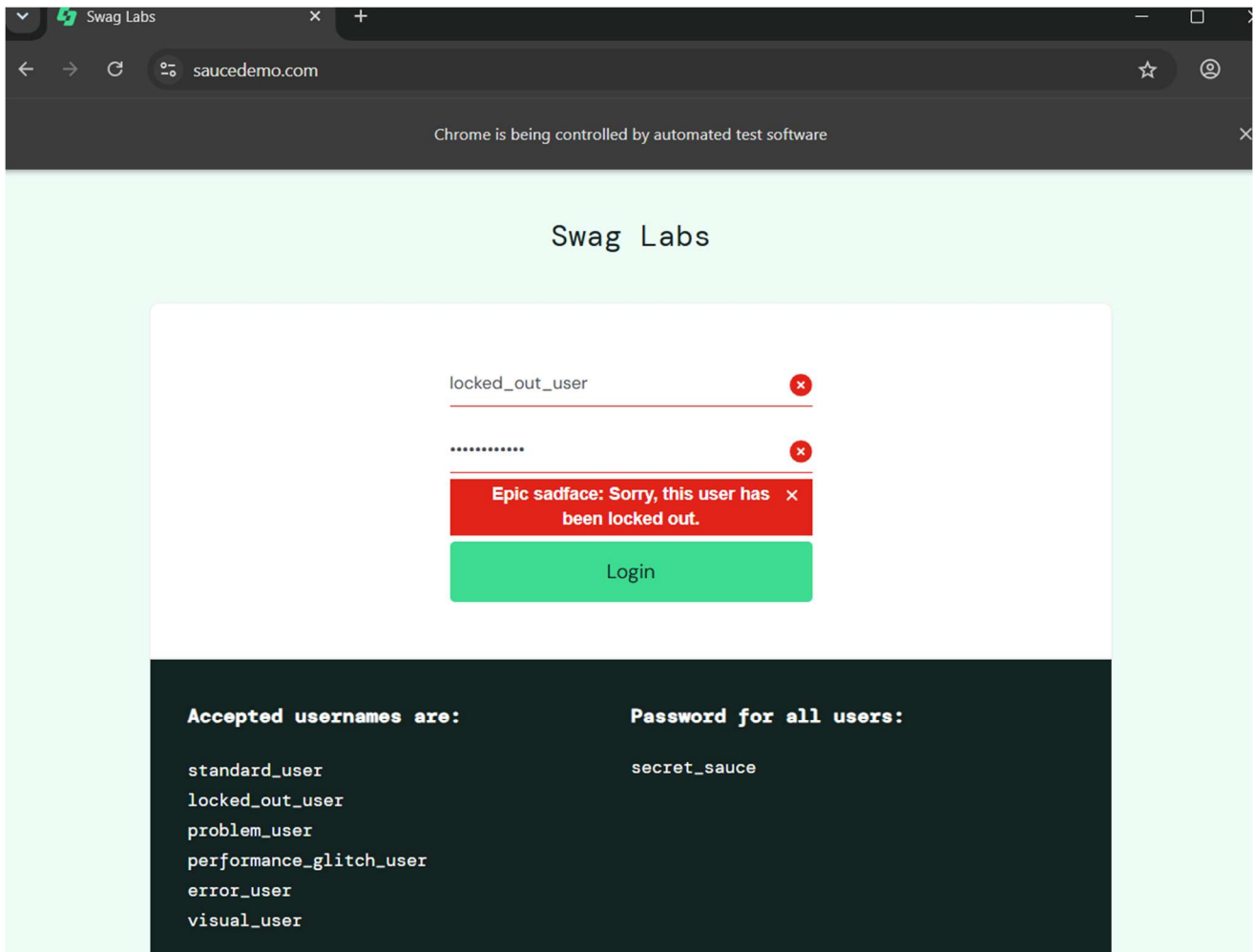
Test data

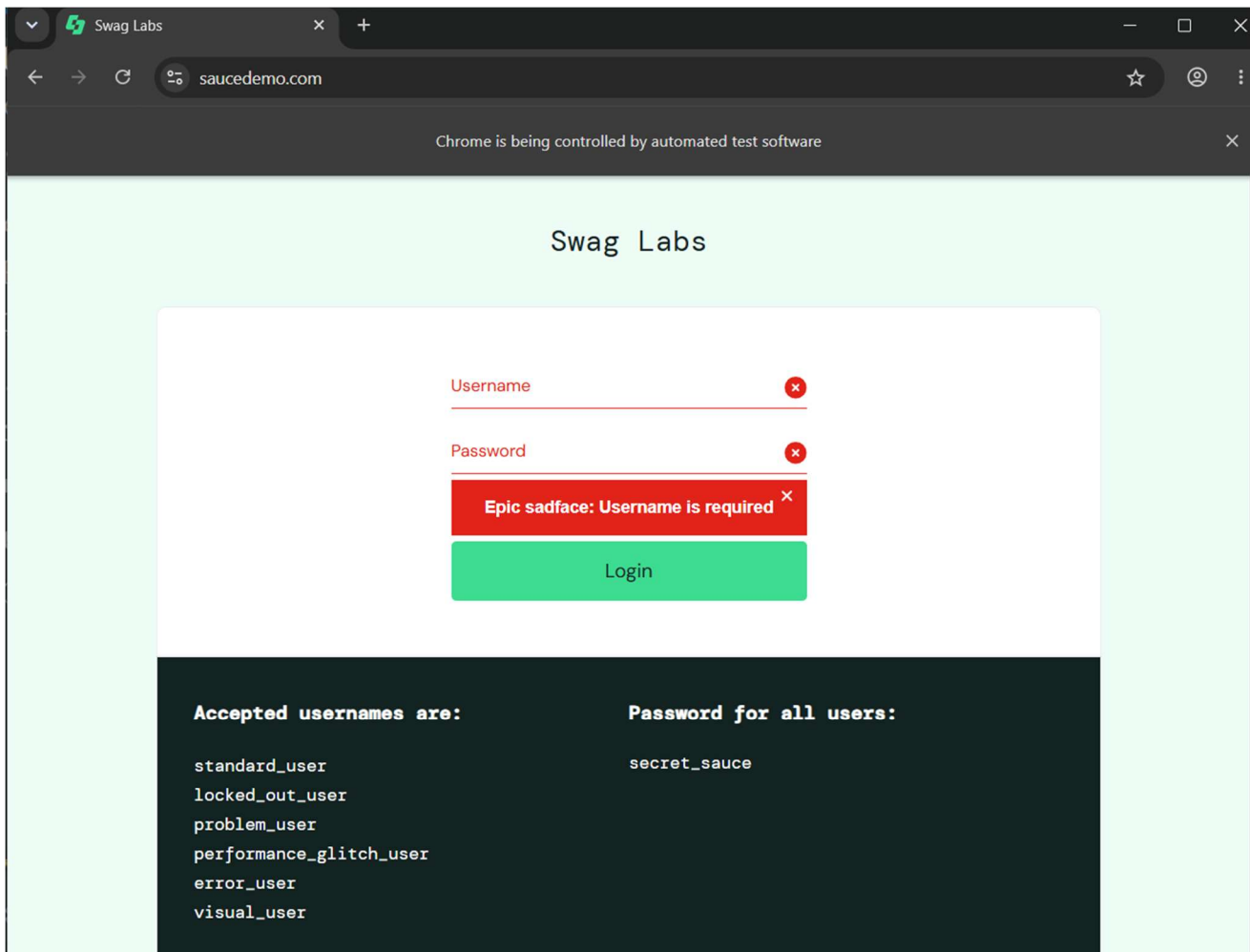
Scenario	Username	Password
Locked out user	locked_out_user	secret_sauce
Unknown user	invalid_user	wrong_pass
Blank credentials	(empty)	(empty)

Running the project

Open the supplied project folder in VS Code, create a Python virtual environment, install the packages from requirements.txt, and run the suite from the integrated terminal:

```
py -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
python -m robot --outputdir results Experiment2.robot
```





```
=====
Experiment2 :: Data-driven invalid-login checks for SauceDemo.
=====
Locked out user                                     | PASS |
-----
Unknown user                                       | PASS |
-----
Blank credentials                                 | PASS |
-----
Experiment2 :: Data-driven invalid-login checks for SauceDemo. | PASS |
3 tests, 3 passed, 0 failed
=====
Output: C:\Users\rahul\Documents\Codex\2026-09-23\rewrite-the-document-in-own-words-2\results\output.xml
Log:    C:\Users\rahul\Documents\Codex\2026-09-23\rewrite-the-document-in-own-words-2\results\log.html
Report: C:\Users\rahul\Documents\Codex\2026-09-23\rewrite-the-document-in-own-words-2\results\report.html
```

Expected result

Robot Framework executes three tests from the one template. Each test uses a different credential row and checks for the invalid-login message. HTML reports are written to the results folder.

Conclusion

A test template makes the automation workflow reusable. To add coverage, a new credential row can be added to the test-case table without copying the browser actions or assertions.